- Rebuilt from my own research project, Play to Learn in VR (P01.01), scaled to a study one graduate student can finish in this course: one coder, about three hours of coding, 150 items.
- Everything under "The example" is what a student would submit. Every outline point is a full sentence that makes a claim.
- The rubric map at the end shows which grading row each section earns and why. Your topic and your data source are your own choice.

The example

Learning Talk and VR Tone: Affordances and Constraints in Tweets About Playing in Virtual Reality

Framing

1. People who play in virtual reality (VR) talk about it in public, and a share of that talk describes learning: picking up a skill in a game, practicing an instrument, training for a job, or teaching someone else how to play.
2. This study examines English-language tweets posted in 2021 that mention VR or virtual reality together with play, an archive of 100,363 original posts collected through the Twitter Academic Research API before that access closed in 2023.
3. Schools and training programs are deciding now whether to buy headsets for play-based learning, and the evidence they weigh comes mostly from lessons researchers designed and from surveys, not from what users say on their own.
4. If learning talk about VR is mostly positive and names what VR lets a person do, buyers can expect a receptive audience and should budget for the obstacles users name; if it is not, then the enthusiasm in the trade press overstates what users experience, and the obstacles deserve the first dollar.

Theoretical positioning

1. Affordance theory holds that a person perceives an environment in terms of the actions it offers, its affordances, and the actions it blocks, its constraints (Gibson, 1979).
2. On this account, a tweet that brings learning into talk about playing in VR is describing VR as an environment that offers learning, so the writer should name what VR lets a person do and should evaluate VR favorably while doing it.

3. The theory therefore predicts that tweets that refer to learning will be more often positive toward VR, and will name affordances more often than constraints, compared with tweets about playing in VR that do not refer to learning.
4. A competing account, the cognitive-load view of immersive learning, holds that the features that make VR playful and immersive also pull attention away from the lesson and add to the effort of learning, so the promise of VR learning is kept only when design holds attention on the task (Mayer et al., 2023).
5. On that account, people who describe learning in a play-centered VR setting should report distraction, discomfort, and effort at least as often as benefits, so learning tweets should be no more positive than other tweets and should carry constraints as often as affordances.
6. Comparing the tone and the affordance-to-constraint balance of learning tweets against non-learning tweets lets the data favor one account over the other, because the two accounts predict opposite differences on the same two variables.

The literature

1. Large-scale studies of public posts find talk about VR in education mostly positive: anticipation, trust, and joy were the dominant emotions in 17 million tweets about augmented and virtual reality (Lampropoulos et al., 2022), and VR was discussed more favorably than augmented or mixed reality in posts about extended reality in higher education (Al-Samarraie et al., 2025).
2. The positive picture is not uniform, because comments under TikTok, YouTube, and Facebook posts about the metaverse in education showed a plurality of neutral sentiment, which the author read as caution rather than enthusiasm (Mahmoud, 2024).
3. Studies that measure emotion directly find that an immersive VR lesson raises enjoyment and engagement relative to video and textbook conditions (Allcoat & von Mühlenen, 2018), and that the emotional value of VR in education rests on presence and control (Makransky & Lilleholt, 2018).
4. The same literature records the constraints that cut VR learning short: cybersickness and accessibility are the persistent barriers across the collaborative-learning studies (van der Meer et al., 2023), motion sickness recurs across the 2019 to 2024 studies of immersive VR in higher education (Sunardi et al., 2025), and perceived usefulness and enjoyment are what keep people using VR when those barriers are present (Jo & Park, 2023).
5. Students asked about collaborative VR name affordances and challenges in comparable numbers, which suggests that the two travel together in how people talk about the technology (Laine & Lee, 2024).

6. Whether play helps or hurts learning in VR is contested: design studies treat playful immersion as a source of engagement, while the cognitive-load account treats the same features as a distraction from the lesson (Mayer et al., 2023).
7. The studies share one limitation: the discourse studies score emotion by counting words from a lexicon such as the NRC word-emotion lexicon (Mohammad & Turney, 2013), which cannot read irony, negation, or an advertisement that uses happy words, and the lesson studies measure feeling after a lesson the researchers designed, so none of them reads unprompted posts by hand for whether learning changes the tone.

The gap

1. Because the discourse studies counted emotion words and did not separate play from VR in general, none of them can say whether tweets about playing in VR that refer to learning differ in tone from those that do not.
2. Because the lesson studies asked participants about a lesson the researchers built, the affordances and constraints people volunteer about playing in VR, in their own words, have not been coded by hand.
3. Both open questions are what this study asks, on a hand-coded sample of the play-in-VR discourse.

Research questions and hypotheses

1. H1: Tweets about playing in VR that refer to learning are more likely to be positive toward VR than tweets about playing in VR that do not refer to learning. (Theory points 1 to 3.)
2. RQ1: Which affordances and which constraints of VR appear most often in tweets that refer to learning? (Theory points 1 and 2.)
3. RQ2: Is the balance of affordances to constraints different in tweets that refer to learning than in tweets that do not? (Theory points 3 to 6; this is the comparison that separates the two accounts.)

The proposed design

1. The unit of analysis is one tweet: an English-language original post, not a retweet, posted between January 1 and December 31, 2021, that mentions VR or virtual reality together with play.
2. The population is the 100,363 such tweets in the archive collected through the Twitter Academic Research API; the frame is that archive, with the collection date recorded, and tweets the platform flagged as possibly sensitive are removed before sampling because the archive contains adult-game advertising that uses the word train.

3. I will draw a stratified random sample of 150 tweets with a fixed seed: 75 from tweets that contain one of the 34 learning words the original project used (learn, teach, train, study, student, and their forms) and 75 from tweets that contain none, so that both groups are present; group membership in the analysis comes from my coding, not from the word list, because the list also catches game titles and idioms.
4. Learning reference records whether the tweet describes a person learning, teaching, studying, practicing, or being trained (0 or 1; nominal).
5. Tone toward VR records the writer's evaluation of VR or of the VR experience (1 negative, 2 neutral or mixed, 3 positive; ordinal).
6. Affordance named records the benefit of VR the tweet names (none; fun or engagement; hands-on interaction; doing it with others; practicing a skill; other; nominal), and Constraint named records the obstacle it names (none; physical discomfort; hardware cost; technical performance; other; nominal).
7. The like count at collection is recorded beside the coded variables as a looked-up number (ratio).
8. Tweets sit on the 75-seconds-each row of the sizing table, so 150 tweets with four coded variables is about three hours of coding, inside the two-to-four-hour budget; I will time the first 20 and rescale the sample if my rate differs.
9. I am the only coder, so I will recode a random 15 tweets, 10 percent of the sample, at least 48 hours after the first pass with the first codes hidden, and report percent agreement for each variable with the disagreements listed; 10 percent is enough to practice the mechanics here, and a study meant for publication would recode 10 to 20 percent.
10. For H1 I will cross-tabulate learning reference against tone and test the association with a chi-square test (a test of whether two sets of categories are related), reporting the effect size and which cells carry the difference.
11. For RQ1 I will report the frequency of each affordance category and each constraint category among tweets coded 1 on learning reference.
12. For RQ2 I will record for each tweet whether it names an affordance only, a constraint only, both, or neither, and cross-tabulate that against learning reference with a chi-square test; a larger affordance share among learning tweets favors affordance theory, and an equal or smaller share favors the cognitive-load account.

Contribution

1. The study tests affordance theory's prediction on what users write unprompted rather than on what they report after a lesson someone designed, which tells researchers of VR in education whether the positive register found by word counting survives a hand coding that can read advertisements, idioms, and irony.

2. It gives educators and designers deciding on VR play a list of the benefits users volunteer and the obstacles users say made them stop, drawn from the users' own posts.
3. It shows discourse researchers whether a learning word list and a hand-coded learning reference pick out the same tweets, which bears on every study that has used a dictionary to define a learning subset.

References

- Al-Samarraie, H., Sarsam, S. M., Alzahrani, A. I., & Aldowah, H. (2025). Exploring sentiments and topics in extended reality learning environments: A comparative study. *PLOS ONE, 20*(7), Article e0327311. https://doi.org/10.1371/journal.pone.0327311
- Allcoat, D., & von Mühlenen, A. (2018). Learning in virtual reality: Effects on performance, emotion and engagement. *Research in Learning Technology, 26*. https://doi.org/10.25304/rlt.v26.2140
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.
- Jo, H., & Park, D.-H. (2023). Affordance, usefulness, enjoyment, and aesthetics in sustaining virtual reality engagement. *Scientific Reports, 13*(1), Article 15097. https://doi.org/10.1038/s41598-023-42113-1
- Laine, T. H., & Lee, W. (2024). Collaborative virtual reality in higher education: Students' perceptions on presence, challenges, affordances, and potential. *IEEE Transactions on Learning Technologies, 17*, 280-293. https://doi.org/10.1109/TLT.2023.3319628
- Lampropoulos, G., Keramopoulos, E., Diamantaras, K., & Evangelidis, G. (2022). Augmented reality and virtual reality in education: Public perspectives, sentiments, attitudes, and discourses. *Education Sciences, 12*(11), Article 798. https://doi.org/10.3390/educsci12110798
- Mahmoud, A. B. (2024). Exploring the public's beliefs, emotions and sentiments towards the adoption of the metaverse in education: A qualitative inquiry using big data. *British Educational Research Journal, 50*(5), 2320-2341. https://doi.org/10.1002/berj.4026
- Makransky, G., & Lilleholt, L. (2018). A structural equation modeling investigation of the emotional value of immersive virtual reality in education. *Educational Technology Research and Development, 66*(5), 1141-1164. https://doi.org/10.1007/s11423-018-9581-2
- Mayer, R. E., Makransky, G., & Parong, J. (2023). The promise and pitfalls of learning in immersive virtual reality. *International Journal of Human-Computer Interaction, 39*(11), 2229-2238. https://doi.org/10.1080/10447318.2022.2108563

- Mohammad, S. M., & Turney, P. D. (2013). Crowdsourcing a word-emotion association lexicon. *Computational Intelligence, 29*(3), 436-465. https://doi.org/10.1111/j.1467-8640.2012.00460.x
- Sunardi, Meyliana, Hendric, S. W. H. L., & Rosmansyah, Y. (2025). Exploring immersive virtual reality in higher education: Research gap and future direction, a scoping review. *IEEE Access, 13*, 76308-76321. https://doi.org/10.1109/ACCESS.2025.3565385
- van der Meer, N., van der Werf, V., Brinkman, W.-P., & Specht, M. (2023). Virtual reality and collaborative learning: A systematic literature review. *Frontiers in Virtual Reality, 4*, Article 1159905. https://doi.org/10.3389/frvir.2023.1159905

Rubric map

Points are the assignments page's own split: framing 8, theory 15, literature 15, gap 8, questions 10, design 15, contribution 4.

Rubric row	Points	Where the example earns it	Why it earns the full row
Framing	8	Framing, points 1 to 4	Four points, inside the three-to-five range. The first point is the phenomenon, not earlier studies. The population is bounded here: platform, query, year, and archive size. Point 3 says why it matters now, and point 4 states the stakes in both directions. No sentence would be true of any topic.
Theoretical positioning	15	Theory, points 1 to 6	One lens is used, not listed: points 1 to 3 give the lens, the reason, and the prediction, with the reason coming from the theory rather than from earlier findings. Points 4 and 5 give a rival that predicts something different in the same data, and point 6 says how the design tells them apart. The points are numbered so the questions can cite them.
The literature	15	Literature, points 1 to 7	Seven points, inside the six-to-ten range, organized by finding rather than by author, with two or more citations behind most points. Point 1 is the convergence, points 2, 5, and 6 are the nuance and the disagreement, and point 7 is the limitation the studies share, which the gap rests on. Nothing repeats the theory section.

Rubric row	Points	Where the example earns it	Why it earns the full row
The gap	8	Gap, points 1 to 3	Both gap points follow from the shared limitation named in literature point 7, name the kind of study they apply to, and state the question left open. Point 3 ties the open questions to this study. No claim that the topic is unstudied.
Research questions or hypotheses	10	H1, RQ1, RQ2	Every item is answerable from coded tweets. H1 links two coded variables in a stated direction as a tendency ("more likely"), claims association rather than cause, and names the theory points it comes from. Each RQ asks one thing, uses only terms the design records, and matches a comparison the design makes.
The proposed design	15	Design, points 1 to 12	Twelve points, inside the eight-to-twelve range, covering all six required parts: one unit for one kind of material; a population and a rebuildable frame with its exclusion rule; a seeded, stratified sampling method; every variable with what it records and its kind of answer; a one-coder reliability plan at 10 percent after 48 hours with first codes hidden; and an analysis named for each question before any data exist. The time budget is tied to a named sizing-table row.
Contribution	4	Contribution, points 1 to 3	Three points for readers in the field (researchers, educators, designers), each claiming only what coded tweets can show. Nothing is promised that no variable measures.
References	required	The reference list	Every source cited in the outline appears once, in APA 7, and nothing uncited is listed.

Length: the example runs about 1,500 words from the title through the contribution section, inside the 1,200-to-1,800-word range. The references are not counted.